

Nathan Tsao

 |  |  | portfolio | US Citizen

Machine learning research engineer with two years of academic research and internship experience in applied machine learning and autonomous systems. Seeking a full-time position in the United States.

EDUCATION

University of Texas at Austin <i>Masters of Science:</i> Mechanical Engineering <i>Thesis:</i> Neural Port-Hamiltonian Differential Algebraic Equations	Aug 2023 – May 2025 GPA: 3.74
University of Illinois Urbana Champaign <i>Bachelors of Science:</i> Mechanical Engineering	Aug 2019 – May 2022 GPA: 3.87

WORK EXPERIENCE

NASA Ames Research Center Autonomous Aircraft Operations Research Intern - Researching how traffic following behavior in dense airspaces effect aircraft performance metrics. - Developing GPU-accelerated multi-modal traffic simulation of the Bay Area to guide infrastructure development of Unmanned Aerial Vehicles.	June 2025 – Aug 2025 Mountain View, CA
Berkeley Lights Hardware Engineering Intern - Developed mechanical and electrical design of temperature calibration sensor for the OptoSelect chip. - Automated data acquisition and calibration pipeline using Python scripts.	June 2022 – Sep 2022 Berkeley, CA

RESEARCH EXPERIENCE

Autonomous Systems Group: UT Austin Graduate Research Assistant - Developed a scalable electrical circuit modeling framework using compositional machine learning models. - Created low-energy and fast-inference human activity recognition algorithm for batteryless sensors.	Dec 2023 – May 2025 Austin, TX
Hybrid Robotics Group: UC Berkeley Visiting Research Assistant - Led team to design the hardware and RL controller design of an actuated tail for quadruped robots. - Created inexpensive 3D-printed quasi-direct-drive actuators with custom FOC motor controller.	May 2022 – Jan 2023 Berkeley, CA
RoboDesign Lab: UIUC Undergraduate Research Assistant Prototyped a low-cost force-sensing humanoid robot foot using elastomers and Hall sensors.	Jan 2022 – May 2022 Urbana, IL

PREPRINTS

Cyrus Neary*, Nathan Tsao*, and Ufuk Topcu. Neural Port-Hamiltonian Differential Algebraic Equations for Compositional Learning of Electrical Networks. *Under review at CDC 2025*.

Geffen Cooper*, Nathan Tsao*, Filippas Fotiadis, Ufuk Topcu, Radu Marculescu. Learning from Sparse and Asynchronous Data Streams for Batteryless Sensors. *Under review at NeurIPS 2025*.

TEACHING EXPERIENCE

ASE 370C Feedback Control Systems: UT Austin Graduate Teaching Assistant	Jan 2025 – May 2025 Austin, TX
ME 314D Dynamics: UT Austin Graduate Teaching Assistant	Sep 2023 – Dec 2023 Austin, TX

HONORS AND AWARDS

Dr. J. Parker Lamb Endowed Presidential Fellowship: UT Austin	2023
Nominated by the Walker Department of Mechanical Engineering.	
Highest Honors: UIUC	2022
Awarded to students with at least a 3.8 Illinois GPA and continual commitment to service and education.	
Dean's List: UIUC	2019-22
Awarded to undergraduate students in the top 20 percent of their college class.	

OUTREACH

Del Valle High School Visit	Mar 2025
Presented machine learning academic research to engineering and robotics high school students.	
UT Austin Girls Day	Feb 2024
Guided K-8th graders to program remote-controlled cars in Python.	

SKILLS

Skills: Deep Learning Generative AI Linux Statistics
Programming Languages: Python Dart
Frameworks: PyTorch Jax Docker Stable-Baselines3 Git
Languages: English Mandarin
Relevant Coursework: Reinforcement Learning Statistical Machine Learning Theoretical Statistics Convex Optimization